

Macroeconomics #3

Ramsey–Cass–Koopmans Model

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Today's lecture

- ① Motivation
- ② Ramsey in discrete time
- ③ Phase diagram analysis
- ④ Ramsey in continuous time

Introduction and Motivation

From Solow to Ramsey

Recall the Solow model:

- The saving rate s is **exogenous** and constant.
- Households save a fixed fraction of income regardless of returns.
- This is a strong assumption — why would agents not respond to incentives?

The **Ramsey–Cass–Koopmans model** (1928, 1965):

- Households **optimally choose** consumption and saving.
- Savings is endogenous, driven by intertemporal optimization.
- Closes the behavioral gap left by Solow's exogenous saving rate.

Key features of the Ramsey model

- Representative household maximizes lifetime utility over an infinite horizon.
- Consumption–saving decision governed by an Euler equation.
- The model delivers the same long-run steady state as Solow (with appropriate s), but with micro-founded transition dynamics.
- Importantly: the Ramsey steady state is always dynamically efficient — no over-accumulation of capital.

We study the model in two formulations:

- ① Discrete time (Lagrangian approach)
- ② Continuous time (Hamiltonian / optimal control)

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Ramsey Model in Discrete Time

Household preferences

A representative household lives forever ($t = 0, 1, 2, \dots$) and maximizes:

$$V_0 = \sum_{t=0}^{\infty} \beta^t U(c_t)$$

where:

- c_t is per-capita consumption at time t .
- $\beta \in (0, 1)$ is the **discount factor** (patience parameter).
- $U(\cdot)$ is the instantaneous utility function.

Assumptions on U :

- $U'(c) > 0$ (more is better).
- $U''(c) < 0$ (diminishing marginal utility / desire to smooth consumption).
- Inada conditions: $\lim_{c \rightarrow 0} U'(c) = \infty$, $\lim_{c \rightarrow \infty} U'(c) = 0$.

The most common specification is **Constant Relative Risk Aversion** (CRRA):

$$U(c) = \frac{c^{1-\sigma} - 1}{1-\sigma}, \quad \sigma > 0$$

- σ is the **coefficient of relative risk aversion**: $-\frac{cU''(c)}{U'(c)} = \sigma$.
- $1/\sigma$ is the **elasticity of intertemporal substitution** (EIS).
- When $\sigma \rightarrow 1$: $U(c) = \ln(c)$.
- High σ : household dislikes consumption fluctuations (strong desire to smooth).
- Low σ : household is more willing to substitute consumption over time.

Technology and resource constraint

Production:

$$Y_t = F(K_t, L_t), \quad y_t = f(k_t) \equiv F(k_t, 1)$$

where f satisfies the neoclassical assumptions (CRS, $f' > 0$, $f'' < 0$, Inada).

In per-capita terms, the **resource constraint** is:

$$c_t + k_{t+1} = f(k_t) + (1 - \delta)k_t$$

- Output is split between consumption and gross investment.
- $\delta \in [0, 1]$ is the depreciation rate.
- Given initial capital $k_0 > 0$.

Investment and capital accumulation

Define net investment:

$$k_{t+1} - k_t = f(k_t) - \delta k_t - c_t$$

- $f(k_t) - \delta k_t$ is net output (after replacing depreciated capital).
- Whatever is not consumed becomes net addition to the capital stock.

Compare with Solow:

- Solow: $c_t = (1 - s)f(k_t)$ with s exogenous.
- Ramsey: c_t is **chosen optimally** by the household.

The planner's problem

$$\max_{\{c_t, k_{t+1}\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t U(c_t)$$

subject to:

$$c_t + k_{t+1} = f(k_t) + (1 - \delta)k_t, \quad \forall t \geq 0$$

with $k_0 > 0$ given and the non-negativity constraints $c_t \geq 0$, $k_{t+1} \geq 0$.

Solution method: Form the Lagrangian.

$$\mathcal{L} = \sum_{t=0}^{\infty} \beta^t \left\{ U(c_t) + \lambda_t [f(k_t) + (1 - \delta)k_t - c_t - k_{t+1}] \right\}$$

First-order conditions:

$$\frac{\partial \mathcal{L}}{\partial c_t} = 0 : \quad U'(c_t) = \lambda_t$$

$$\frac{\partial \mathcal{L}}{\partial k_{t+1}} = 0 : \quad \lambda_t = \beta \lambda_{t+1} [f'(k_{t+1}) + 1 - \delta]$$

The multiplier $\lambda_t = U'(c_t)$ is the **shadow price** of capital at time t .
By the Envelope theorem

$$\lambda_{t+1} = U'(c_{t+1})$$

The Euler equation (discrete time)

Using the last three equations:

$$U'(c_t) = \beta U'(c_{t+1})[f'(k_{t+1}) + 1 - \delta]$$

Interpretation:

- **LHS**: Marginal utility cost of saving one more unit today.
- **RHS**: Discounted marginal utility gain from the extra return tomorrow.
- At the optimum, the household is **indifferent** between consuming today and saving for tomorrow.

Equivalently:

$$\frac{U'(c_t)}{\beta U'(c_{t+1})} = f'(k_{t+1}) + 1 - \delta \equiv R_{t+1}$$

The MRS equals the gross return on capital.

Euler equation: consumption dynamics

With CRRA utility $U(c) = \frac{c^{1-\sigma}-1}{1-\sigma}$, the Euler equation becomes:

$$c_t^{-\sigma} = \beta c_{t+1}^{-\sigma} [f'(k_{t+1}) + 1 - \delta]$$

Rearranging:

$$\frac{c_{t+1}}{c_t} = \left\{ \beta [f'(k_{t+1}) + 1 - \delta] \right\}^{1/\sigma}$$

- Consumption **grows** when the net return exceeds the discount rate: $f'(k_{t+1}) + 1 - \delta > 1/\beta$.
- Consumption **falls** when returns are low: $f'(k_{t+1}) + 1 - \delta < 1/\beta$.
- Higher $\sigma \Rightarrow$ less responsive consumption growth (stronger smoothing motive).

Transversality condition (TVC)

The Euler equation alone admits multiple solutions. We need a **boundary condition** at infinity:

$$\lim_{t \rightarrow \infty} \beta^t U'(c_t) k_{t+1} = 0$$

Interpretation:

- The present value of capital left at the end of time must be zero.
- If $\lim \beta^t U'(c_t) k_{t+1} > 0$: the household is “dying rich” — it could increase utility by consuming more.
- The TVC rules out paths where capital grows too fast (inefficiently).

Together, the **Euler equation + resource constraint + TVC** fully characterize the optimal path $\{c_t, k_t\}_{t=0}^{\infty}$.

Optimal steady state

At the steady state, $c_t = c_{t+1} = c^{ss}$ and $k_t = k_{t+1} = k^{ss}$. The Euler equation gives:

$$1 = \beta[f'(k^{ss}) + 1 - \delta]$$
$$\Rightarrow f'(k^{ss}) = \frac{1}{\beta} - 1 + \delta = \delta + \theta$$

where $\theta \equiv \frac{1-\beta}{\beta}$ is the **rate of time preference**.

Steady-state consumption:

$$c^{ss} = f(k^{ss}) - \delta k^{ss}$$

Comparative statics of the steady state

From $f'(k^{ss}) = \delta + \theta$:

- **More patient** ($\beta \uparrow$, i.e. $\theta \downarrow$): $f'(k^{ss})$ falls $\Rightarrow k^{ss} \uparrow \Rightarrow c^{ss} \uparrow$.
 - Patient households save more, accumulate more capital.
- **Higher depreciation** ($\delta \uparrow$): $f'(k^{ss})$ rises $\Rightarrow k^{ss} \downarrow \Rightarrow c^{ss} \downarrow$.
- **Higher TFP** ($z \uparrow$ in $f(k) = zk^\alpha$): $k^{ss} \uparrow \Rightarrow c^{ss} \uparrow$.

Note: the risk aversion parameter σ does **not** affect the steady state — it only affects the **speed of transition**.

Parametric example: CRRA + Cobb–Douglas

Let $U(c) = \frac{c^{1-\sigma}-1}{1-\sigma}$ and $y = zk^\alpha$. Then:

$$\text{Euler: } c_t^{-\sigma} = \beta c_{t+1}^{-\sigma} [1 + \alpha z k_{t+1}^{\alpha-1} - \delta]$$

$$\text{Resource: } k_{t+1} = zk_t^\alpha + (1 - \delta)k_t - c_t$$

Steady state:

$$k^{ss} = \left(\frac{\alpha z}{\delta + \theta} \right)^{\frac{1}{1-\alpha}}, \quad c^{ss} = z(k^{ss})^\alpha - \delta k^{ss}$$

Log-linear + Cobb–Douglas

Let $U(c) = \ln c$ and $y = zk^\alpha$. Then:

$$\text{Euler: } c_{t+1} = \beta c_t \left[1 + \alpha z k_{t+1}^{\alpha-1} - \delta \right]$$

$$\text{Resource: } k_{t+1} = zk_t^\alpha + (1 - \delta)k_t - c_t$$

Steady state:

$$k^{ss} = \left(\frac{\alpha z}{\delta + \theta} \right)^{\frac{1}{1-\alpha}}, \quad c^{ss} = z(k^{ss})^\alpha - \delta k^{ss}$$

where $\beta = 1/(1 + \theta)$.

Dynamics and Phase Diagram (Discrete Time)

The model is summarized by a system

$$\begin{aligned}U'(c_{t+1})[f'(k_{t+1} + 1 - \delta)] &= \frac{1}{\beta}U'(c_t), \\k_{t+1} &= f(k_t) + (1 - \delta)k_t - c_t\end{aligned}$$

For example, when utility is log-linear

$$\begin{aligned}c_{t+1} &= \beta c_t (f'(k_t) + 1 - \delta), \\k_{t+1} &= f(k_t) + (1 - \delta)k_t - c_t\end{aligned}$$

Express in deviation

$$\begin{aligned}\Delta c_t &= c_{t+1} - c_t \\ \Delta k_t &= k_{t+1} - k_t\end{aligned}$$

Phase diagram: the $\Delta k = 0$ locus

From the resource constraint:

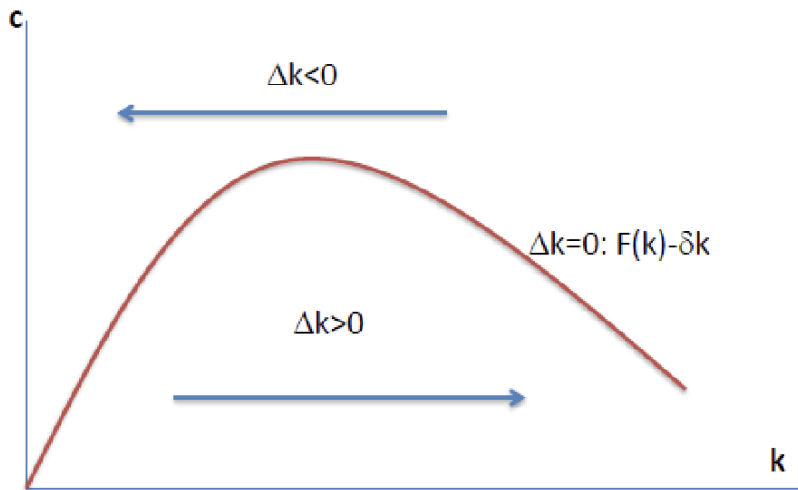
$$\Delta k \equiv k_{t+1} - k_t = f(k_t) - \delta k_t - c_t$$

The $\Delta k = 0$ locus is:

$$c = f(k) - \delta k$$

changes

- This is a hump-shaped curve (why?).
- **Above** the locus: $c > f(k) - \delta k \Rightarrow \Delta k < 0$ (capital falls).
- **Below** the locus: $c < f(k) - \delta k \Rightarrow \Delta k > 0$ (capital rises).



Dynamics of capital

Phase diagram: the $\Delta c = 0$ locus

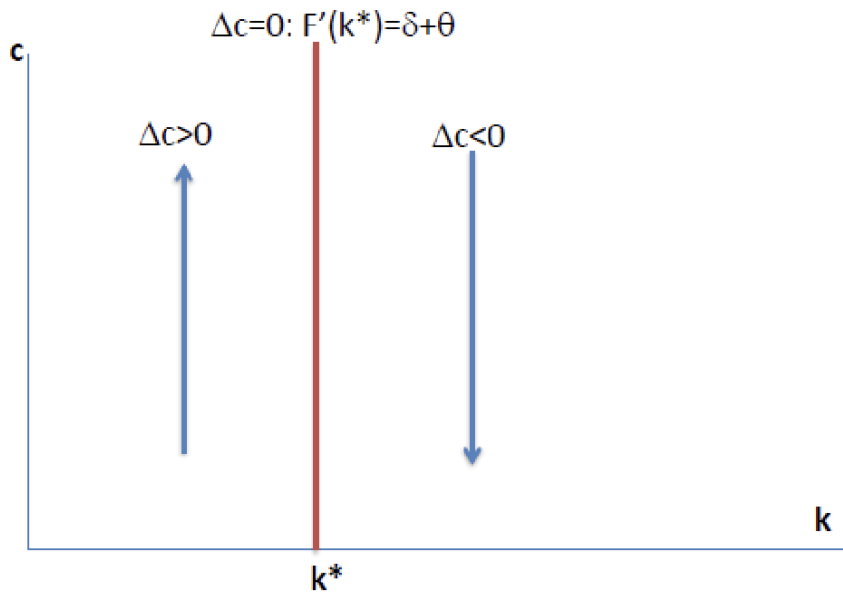
From the Euler equation, $c_{t+1} = c_t$ when:

$$\beta[f'(k) + 1 - \delta] = 1 \quad \Rightarrow \quad f'(k^*) = \delta + \theta$$

The $\Delta c = 0$ locus is a **vertical line** at $k = k^*$.

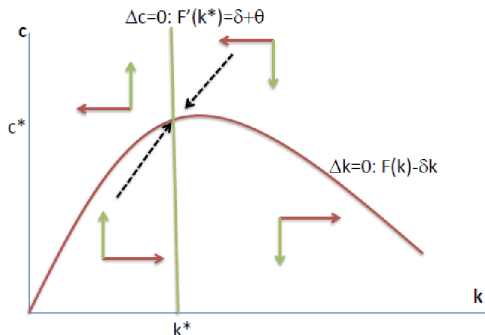
Pay attention to the sign of Δc when k changes

- If $k_{t+1} < k^*$: $f'(k_{t+1}) > \delta + \theta \Rightarrow$ returns are high $\Rightarrow \Delta c > 0$ (consumption rising).
- If $k_{t+1} > k^*$: $f'(k_{t+1}) < \delta + \theta \Rightarrow$ returns are low $\Rightarrow \Delta c < 0$ (consumption falling).



Dynamics of consumption

Putting together: the phase diagram



Phase diagram: dynamics of consumption and capital

- The two loci divide the (k, c) plane into four regions.
- In each region, arrows indicate the direction of motion.
- The steady state (k^*, c^*) is at the intersection.

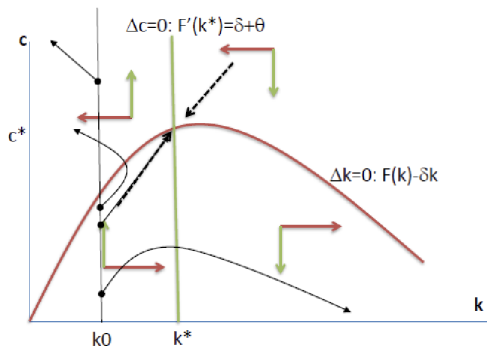
Dynamics of the model

- From the phase diagram, given any initial (k_0, c_0) , we can trace the dynamics of $\{k_t, c_t\}$.
- But given k_0 , the dynamics from an **arbitrary** c_0 will **not** converge to the steady state.
- However, given k_0 , there is a **unique** value of c_0 such that the economy converges to (k^*, c^*) .
- This unique trajectory is called the **saddle path**.

The optimal solution lies on the saddle path:

- c_0 is a function of k_0 : $c_0 = c_0(k_0)$.
- Paths above the saddle path lead to $k \rightarrow 0$ (capital exhaustion).
- Paths below the saddle path violate the TVC (over-accumulation).

The saddle path in the phase diagram

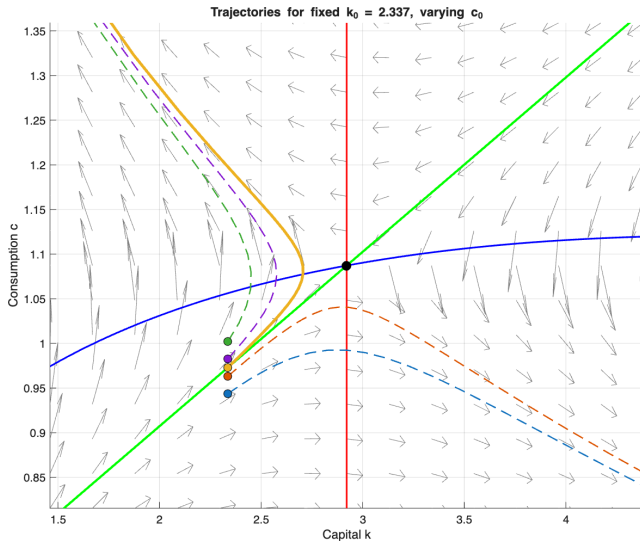


Phase diagram: saddle path and dynamics from different initial consumption levels

- k is **predetermined** (state variable).
- c **jumps** to the saddle path at $t = 0$ (control variable).
- The economy then follows the saddle path to the steady state.

Key Points: Saddle Path in the Ramsey Model

- **Saddle-path stability:** The system has exactly one stable eigenvalue and one unstable eigenvalue. The stable manifold (the **saddle path**) is one-dimensional.
- For a given initial capital k_0 , **there exists a unique consumption** c_0 such that the resulting path converges to the steady state. That unique c_0 is the point on the saddle path above k_0 .
- If you choose any other c_0 , the path will eventually violate the transversality condition:
 - Capital goes to zero (consumption collapses), or
 - Consumption grows without bound while capital becomes negative.Both are infeasible or suboptimal in the Ramsey model.
- In the phase diagram, trajectories starting **exactly on** the green line (the linearised saddle path) converge to the steady state. Those starting slightly **off** it diverge visibly because the unstable component grows exponentially.



Comparative Dynamics

Comparative dynamics: increase in θ (less patience)

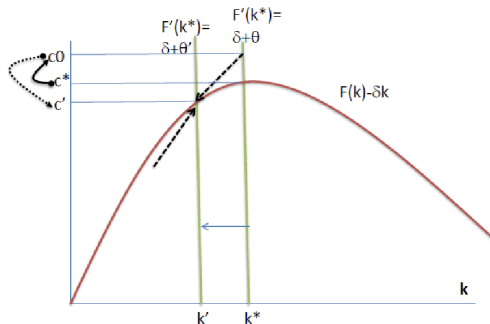
Suppose households become **less patient**: θ increases (β falls).

- The $\Delta c = 0$ locus: $f'(k^*) = \delta + \theta$ shifts **to the left** (lower k^*).
- The $\Delta k = 0$ locus: $c = f(k) - \delta k$ is **unaffected**.

Dynamics:

- 1 When θ increases, c_t **jumps up** to the new saddle path.
- 2 In the short run: higher consumption, lower investment.
- 3 In the long run: capital falls to the new (lower) k^* , and consumption falls to the new (lower) c^* .

Phase diagram: increase in θ



Phase diagram: An increase in the discount rate (θ).

- Short run: c_0 jumps up $\Rightarrow$ saving falls.
- Transition: k and c both decline along the new saddle path.
- Long run: lower k^* , lower c^* .

Comparative dynamics: improvement in technology

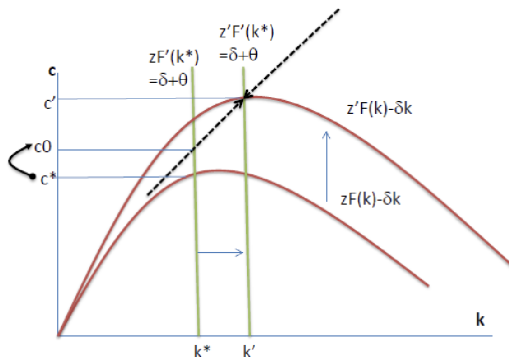
Suppose TFP increases: $z \uparrow$ (so $f(k) = zk^\alpha$ shifts up).

- $\Delta c = 0$: $zf'(k^*) = \delta + \theta \Rightarrow$ shifts **to the right** (higher k^*).
- $\Delta k = 0$: $c = zf(k) - \delta k$ shifts **up** (higher net output).
- In the long run: both k^* and c^* increase.

But what happens in the short run? It depends on σ :

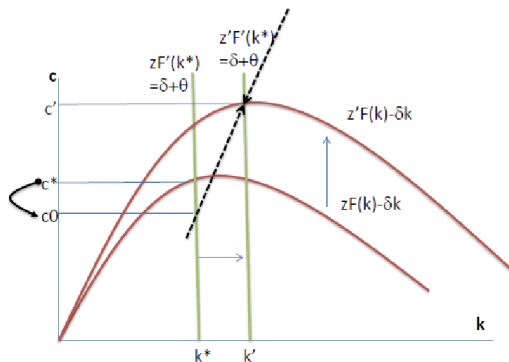
- **High σ** (low EIS, strong smoothing): c_0 jumps up; slow transition.
- **Low σ** (high EIS, willing to substitute): c_0 may fall; fast capital accumulation.

Technology shock: impatient household (high σ)



- High σ , low β , household is impatient and value early periods highly.
- Household wants to smooth consumption $\Rightarrow c_0$ increases immediately.
- Capital accumulates slowly to the new steady state.

Technology shock: patient household (low σ)



- Low σ , high β , household is relatively more patient and value future consumption higher.
- Households want to forgo consumption, invest more to reach the new steady state more quickly.
- c_0 may actually **decrease** in the short run.

Ramsey Model in Continuous Time

Continuous-time setup: social planner

The social planner maximizes the representative household's utility:

$$\max_{\{c(t)\}_{t=0}^{\infty}} \int_0^{\infty} e^{-(\rho-n)t} u(c(t)) dt$$

where:

- $\rho > 0$ is the **rate of time preference** (continuous-time analogue of θ).
- $n \geq 0$ is the population growth rate.
- $\rho - n$ is the **effective discount rate**.
- **Assumption:** $\rho > n$ (effective discounting is positive).

Resource constraint (per capita capital $k(t)$):

$$\dot{k}(t) = f(k(t)) - c(t) - (\delta + n) k(t)$$

with $k(0) > 0$ given. $f(k)$ is the intensive production function.

Continuous-time setup: technology

Aggregate production:

$$Y(t) = F(K(t), L(t), A)$$

Per capita: $y(t) = f(k(t))$ where $k(t) = K(t)/L(t)$. Capital depreciates at rate $\delta \geq 0$. Population grows at rate n . The change in capital per capita is given by the resource constraint:

$$\dot{k}(t) = f(k(t)) - c(t) - (\delta + n) k(t)$$

No separate firm optimization is needed; the planner directly allocates resources.

The Hamiltonian

$$H = u(c(t)) e^{-(\rho-n)t} + \lambda(t) [f(k(t)) - c(t) - (\delta + n) k(t)]$$

- $k(t)$: state variable (capital per capita).
- $c(t)$: control variable (consumption).
- $\lambda(t)$: costate variable (shadow price of capital).

Pontryagin's necessary conditions

The optimality conditions (Pontryagin's Maximum Principle):

$$\frac{\partial H}{\partial c(t)} = 0 \quad \Rightarrow \quad u'(c(t)) e^{-(\rho-n)t} = \lambda(t)$$

$$\frac{\partial H}{\partial k(t)} = -\dot{\lambda}(t) \quad \Rightarrow \quad f'(k(t)) - (\delta + n) = -\frac{\dot{\lambda}(t)}{\lambda(t)}$$

$$\frac{\partial H}{\partial \lambda(t)} = \dot{k}(t)$$

Transversality condition:

$$\lim_{t \rightarrow +\infty} \lambda(t) k(t) = 0$$

Deriving the Keynes–Ramsey rule

From the first FOC: $\lambda(t) = u'(c(t)) e^{-(\rho-n)t}$.

Differentiate both sides with respect to time:

$$\frac{\dot{\lambda}(t)}{\lambda(t)} = \frac{\dot{c}(t) u''(c(t))}{u'(c(t))} + n - \rho$$

Combine with the second FOC ($f'(k) - (\delta + n) = -\dot{\lambda}/\lambda$):

$$f'(k) - (\delta + n) = \rho - n - \frac{\dot{c}(t) u''(c(t))}{u'(c(t))}$$

Define the **elasticity of marginal utility**: $\varepsilon_u(c) = -\frac{c u''(c)}{u'(c)}$.

$$\frac{\dot{c}(t)}{c(t)} = \frac{f'(k(t)) - \delta - \rho}{\varepsilon_u(c(t))}$$

This is the **Keynes–Ramsey rule** (consumption Euler equation).

Interpretation of the Keynes–Ramsey rule

$$\frac{\dot{c}}{c} = \frac{f'(k) - \delta - \rho}{\varepsilon_u(c)} \quad \Longleftrightarrow \quad f'(k) - \delta = \rho + \frac{\dot{c}}{c} \varepsilon_u(c)$$

- $f'(k) - \delta$: marginal benefit from saving (net return on capital).
- $\rho + \frac{\dot{c}}{c} \varepsilon_u(c)$: marginal (opportunity) cost of saving, where:
 - ρ : time preference (prefer consumption today).
 - $\frac{\dot{c}}{c} \varepsilon_u(c)$: change in marginal utility due to changing consumption.
- At the optimum, the planner equates the benefit and cost of saving.

With CRRA utility $u(c) = \frac{c^{1-\sigma}-1}{1-\sigma}$:

- $\varepsilon_u(c) = \sigma$ (constant).

The Euler equation simplifies to:

$$\frac{\dot{c}}{c} = \frac{1}{\sigma} [f'(k) - \delta - \rho]$$

- Consumption grows when $f'(k) > \delta + \rho$ (high marginal product).
- Consumption falls when $f'(k) < \delta + \rho$.
- At steady state: $f'(k^*) = \delta + \rho$.

Equilibrium and Dynamics (Continuous Time)

The equilibrium is characterized by two differential equations:

$$\frac{\dot{c}}{c} = \frac{f'(k) - \delta - \rho}{\varepsilon_u(c)} \quad (\text{Euler})$$

$$\dot{k} = f(k) - c - (\delta + n)k \quad (\text{Capital})$$

with initial condition $k(0) > 0$ and the transversality condition.

Capital accumulation in both models:

$$\dot{k} = f(k) - c - (\delta + n)k$$

- **Solow:** $c = (1 - s)f(k) \Rightarrow \dot{k} = sf(k) - (\delta + n)k$, with s **exogenous**.
- **Ramsey:** c is determined by the **Euler equation**. Saving $f(k) - c$ is **endogenous**.

The key difference: in Ramsey, the saving decision is determined by the planner's optimality condition (Euler equation), not imposed exogenously.

Steady state (continuous time)

At the steady state, $\dot{c} = 0$ and $\dot{k} = 0$:

$$\dot{c} = 0 : \quad f'(k^*) = \delta + \rho$$

$$\dot{k} = 0 : \quad c^* = f(k^*) - (\delta + n)k^*$$

Proposition. *There exists a unique steady state $(c^*, k^*) > (0, 0)$.*

- k^* is uniquely determined by $f'(k^*) = \delta + \rho$ (since $f'' < 0$).
- c^* is then determined from the resource constraint.
- Under $\rho > n$, the TVC is satisfied at steady state.

Modified Golden Rule

The condition $f'(k^*) = \delta + \rho$ is called the **Modified Golden Rule**.

Recall the (Solow) Golden Rule: $f'(k_g^{gold}) = \delta + n$.

- Since $\rho > n$: $f'(k^*) = \delta + \rho > \delta + n = f'(k_g^{gold})$.
- Therefore $k^* < k_g^{gold}$.
- The Ramsey steady state has **less capital** than the Golden Rule.

Why?

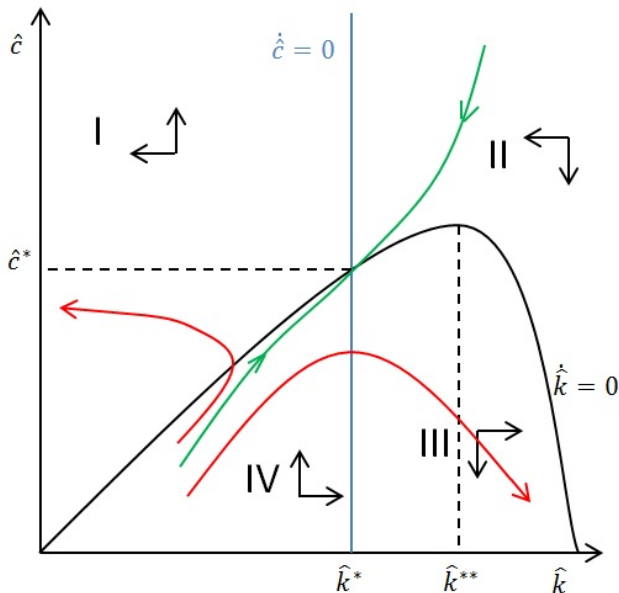
- Households prefer earlier consumption (discount factor $\rho > 0$).
- They maximize the **whole path** of consumption, not just the steady-state level.
- The economy is always **dynamically efficient** — no over-accumulation.

The $\dot{c} = 0$ and $\dot{k} = 0$ loci

- **C-locus** ($\dot{c} = 0$): $\{(k, c) : f'(k) = \delta + \rho\}$ — a **vertical line** at $k = k^*$.
 - Left of k^* : $f'(k) > \delta + \rho \Rightarrow \dot{c} > 0$.
 - Right of k^* : $f'(k) < \delta + \rho \Rightarrow \dot{c} < 0$.
- **K-locus** ($\dot{k} = 0$): $c = f(k) - (\delta + n)k$ — a **hump-shaped** curve.
 - Below the curve: $\dot{k} > 0$.
 - Above the curve: $\dot{k} < 0$.

The C-locus and K-locus divide the k - c plane into 4 regions with distinct dynamics.

Four regions of the phase diagram



The saddle path stability can be shown

Linearize the system around (k^*, c^*) :

$$\begin{pmatrix} \dot{k} \\ \dot{c} \end{pmatrix} = J \begin{pmatrix} k - k^* \\ c - c^* \end{pmatrix}, \quad J = \begin{pmatrix} f'(k^*) - \delta - n & -1 \\ \frac{c^*}{\sigma} f''(k^*) & 0 \end{pmatrix}$$

- $\det(J) = \frac{c^*}{\sigma} f''(k^*) < 0$ (since $f'' < 0$). For a formal proof, check [My Note \(p.78\)](#)
- Negative determinant $\Rightarrow$ one **positive** and one **negative** eigenvalue.
- This confirms **saddle-path stability**: one stable dimension, one unstable.
- Since k is predetermined and c can jump, the system has exactly enough “free” variables to place the economy on the stable manifold.

Comparisons

Side-by-side comparison

	Discrete Time	Continuous Time
Objective	$\sum_{t=0}^{\infty} \beta^t U(c_t)$	$\int_0^{\infty} e^{-\rho t} u(c) dt$
Discount	$\beta = \frac{1}{1+\theta}$	$e^{-\rho t}, \rho \approx \theta$
Method	Lagrangian	Hamiltonian
Euler eq.	$U'(c_t) = \beta U'(c_{t+1}) R_{t+1}$	$\dot{c} = \frac{r-\rho}{\varepsilon_u(c)}$
Steady state	$f'(k^*) = \delta + \theta$	$f'(k^*) = \delta + \rho$
TVC	$\lim_{t \rightarrow \infty} \beta^t U'(c_t) k_{t+1} = 0$	$\lim_{t \rightarrow \infty} e^{-\rho t} u'(c) k = 0$
Stability	Saddle path	Saddle path

Both formulations deliver the same economic insights; they differ in mathematical technique.

When to use which?

- Discrete time:
 - Natural for calibration and numerical simulation.
 - Matches data frequency (quarterly, annual).
 - Easier to code (iterate forward).
 - Standard in DSGE / quantitative macro.
- Continuous time:
 - Easier to derive analytical results
 - Phase diagrams are more natural.
 - More applicable in high frequency data

What does Ramsey add to Solow?

	Solow	Ramsey
Saving rate	Exogenous s	Endogenous
Micro-foundations	No	Yes
Dynamic efficiency	Not guaranteed	Always
Golden Rule	Possible to overshoot	No overshooting
Transition dynamics	Monotone convergence	Saddle-path convergence

- Ramsey's main contribution: **endogenous saving** and **dynamic efficiency**.
- Both models share the same limitation: no endogenous long-run growth.

Dynamic efficiency and over-accumulation

- In Solow, if s is “too high,” the economy can end up with $k > k^{gold}$ — **dynamic inefficiency** (everyone could be made better off by consuming more).
- In Ramsey, this **never happens**:

$$f'(k^*) = \delta + \rho > \delta + n = f'(k_g^{gold}) \quad \Rightarrow \quad k^* < k_g^{gold}$$

- Rational, forward-looking agents would never over-accumulate capital because they could increase utility by consuming more today.
- This is a direct consequence of the TVC.

Key takeaways

- 1 The Ramsey model provides **micro-founded saving** and growth transition dynamics.
- 2 The core object is the **Euler equation**, linking consumption growth to the rate of return on capital.
- 3 The steady state satisfies the **Modified Golden Rule**:
 $f'(k^*) = \delta + \rho$ (or $\delta + \theta$ in discrete time).
- 4 The economy is always **dynamically efficient**: $k^* < k^{gold}$.
- 5 **Saddle-path stability**: given initial capital, there is a unique convergent path.
- 6 The TVC ensures agents do not over-accumulate or leave wealth unused.
- 7 Both discrete- and continuous-time formulations yield the same insights.

Exercises

Exercise 1: Discrete-time Euler equation

Consider the planner's problem with $U(c) = \ln(c)$ and $f(k) = k^\alpha$.

- 1 Write down the Lagrangian.
- 2 Derive the Euler equation.
- 3 Find the steady-state k^{ss} and c^{ss} as functions of α , β , and δ .
- 4 How does k^{ss} change when β increases? Interpret.

► Solution

Exercise 2: Continuous-time Hamiltonian

Consider the continuous-time Ramsey model with $u(c) = \frac{c^{1-\sigma}-1}{1-\sigma}$ and $f(k) = Ak^\alpha$.

- 1 Write the current-value Hamiltonian.
- 2 Derive the Euler equation $\dot{c}/c$.
- 3 Find the steady state (k^*, c^*) .
- 4 Verify that $k^* < k^{gold}$ when $\rho > n$.

► Solution

Exercise 3: Phase diagram analysis

Consider the continuous-time system with $f(k) = k^\alpha$, $\alpha = 0.3$, $\delta = 0.05$, $\rho = 0.03$, $n = 0.01$, $\sigma = 2$.

- 1 Compute k^* , c^* , and k_g^{gold} .
- 2 Sketch the phase diagram with the $\dot{c} = 0$ and $\dot{k} = 0$ loci.
- 3 Starting from $k_0 = k^*/2$, describe the transition dynamics qualitatively.
- 4 What happens if ρ increases permanently? Sketch the new phase diagram.

► Solution

Solution to Exercise 1

With $U(c) = \ln c$ and $f(k) = k^\alpha$:

① $\mathcal{L} = \sum_{t=0}^{\infty} \beta^t \{ \ln c_t + \lambda_t [k_t^\alpha + (1 - \delta)k_t - c_t - k_{t+1}] \}.$

② FOCs: $1/c_t = \lambda_t$ and $\lambda_t = \beta \lambda_{t+1} [\alpha k_{t+1}^{\alpha-1} + 1 - \delta].$

Euler equation: $\frac{1}{c_t} = \frac{\beta}{c_{t+1}} [\alpha k_{t+1}^{\alpha-1} + 1 - \delta].$

③ At steady state: $\alpha(k^{ss})^{\alpha-1} = 1/\beta - 1 + \delta$, so

$$k^{ss} = \left(\frac{\alpha}{1/\beta - 1 + \delta} \right)^{1/(1-\alpha)}, \quad c^{ss} = (k^{ss})^\alpha - \delta k^{ss}.$$

④ $\beta \uparrow \Rightarrow \text{denominator} \downarrow \Rightarrow k^{ss} \uparrow$. More patient households save more.

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Solution to Exercise 2

With $u(c) = \frac{c^{1-\sigma}-1}{1-\sigma}$ and $f(k) = Ak^\alpha$:

① Current-value Hamiltonian: $\hat{H} = \frac{c^{1-\sigma}-1}{1-\sigma} + \mu[Ak^\alpha - c - (\delta + n)k]$.

② FOCs: $c^{-\sigma} = \mu$ and $\dot{\mu} = \mu[\rho + \delta + n - \alpha Ak^{\alpha-1}]$.

Euler: $\frac{\dot{c}}{c} = \frac{1}{\sigma}[\alpha Ak^{\alpha-1} - \delta - \rho]$.

③ $\dot{c} = 0$: $k^* = \left(\frac{\alpha A}{\delta + \rho}\right)^{1/(1-\alpha)}$. $\dot{k} = 0$: $c^* = A(k^*)^\alpha - (\delta + n)k^*$.

④ $f'(k_g^{gold}) = \delta + n < \delta + \rho = f'(k^*)$ since $\rho > n$. Since $f'' < 0$, $k^* < k_g^{gold}$.
✓

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Solution to Exercise 3

With $\alpha = 0.3$, $\delta = 0.05$, $\rho = 0.03$, $n = 0.01$:

- ① $k^* = (0.3/0.08)^{1/0.7} = 3.75^{10/7} \approx 6.16$.
 $c^* = (k^*)^{0.3} - 0.06 \times k^* \approx 1.71 - 0.37 = 1.34$.
 $k_g^{gold} = (0.3/0.06)^{10/7} = 5^{10/7} \approx 9.52 > k^*$. ✓
- ② Vertical $\dot{c} = 0$ at k^* ; hump-shaped $\dot{k} = 0$ curve; saddle path through (k^*, c^*) .
- ③ From $k_0 < k^*$: c jumps onto saddle path, then both c and k increase monotonically to (k^*, c^*) .
- ④ $\rho \uparrow$: $\dot{c} = 0$ shifts left, $\dot{k} = 0$ unchanged. c jumps up initially, then both c and k decline to new lower steady state.

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